

Catfish — Date Mode PRD

Version: 0.2.1 (final draft for Phase A handoff) **Status:** Pre-implementation — awaiting paper-prototype validation of clue graph **Owner:** Johnny Clem **Last updated:** April 25, 2026

Changelog

v0.2 → v0.2.1:

- New Section 4.1: Saint Mask three-layer flavor system (visual + social + writing)
- Audio pitch-shift explicitly rejected in Section 4 decisions table
- New Epic 9: Saint Mask asset & content production
- Epic 5 biography template gains `speech_register` field
- `variableOverrides` pattern extended to social-post captions

v0.1 → v0.2:

- Beat JSON schema rewritten to use the `variants[].condition` pattern (Epic 4.1)
- On-demand voice cache moved to `Library/Caches` with 7-day TTL (Epic 2.6)
- Emotion-query mechanic clarified as texture-only, not clue authority (Epic 6.3)
- Tell model locked: cumulative-under-the-hood, binary-in-UI (Section 4)
- New Epic 7.7 — corgi-themed friend chat presentation
- VBR M4A added to bundle-size mitigation (Epic 8.1)

1. Overview

Date Mode is the high-stakes, high-immersion gameplay surface of Catfish. Where App Mode is the *investigation* (chat, swipe, journal), Date Mode is the *confrontation* — the player meets a match in person, hears their voice for the first time, and either gathers the critical clue that breaks the case or walks away unsettled.

Date Mode is where the killer tell most often lands. It is also where the tension between Eastworld-driven ambient dialogue and ElevenLabs-voiced scripted beats becomes visible

to the player. The handoff between those two modes — what we call the **Focus Shift** — is the signature interaction of the game.

The mystery's core fiction is *Whodunnit Among Five*: five distinct, appealing matches, any of whom could be the killer. The player's job is to look closer at the person they'd never suspect. Layered on top is the **Saint Mask** flavor system (Section 4.1) — subliminal cues that the killer's visual presentation, social feed, and conversational register all read as *just slightly too composed*, rewarding observant players and replays.

2. Goals & non-goals

Goals

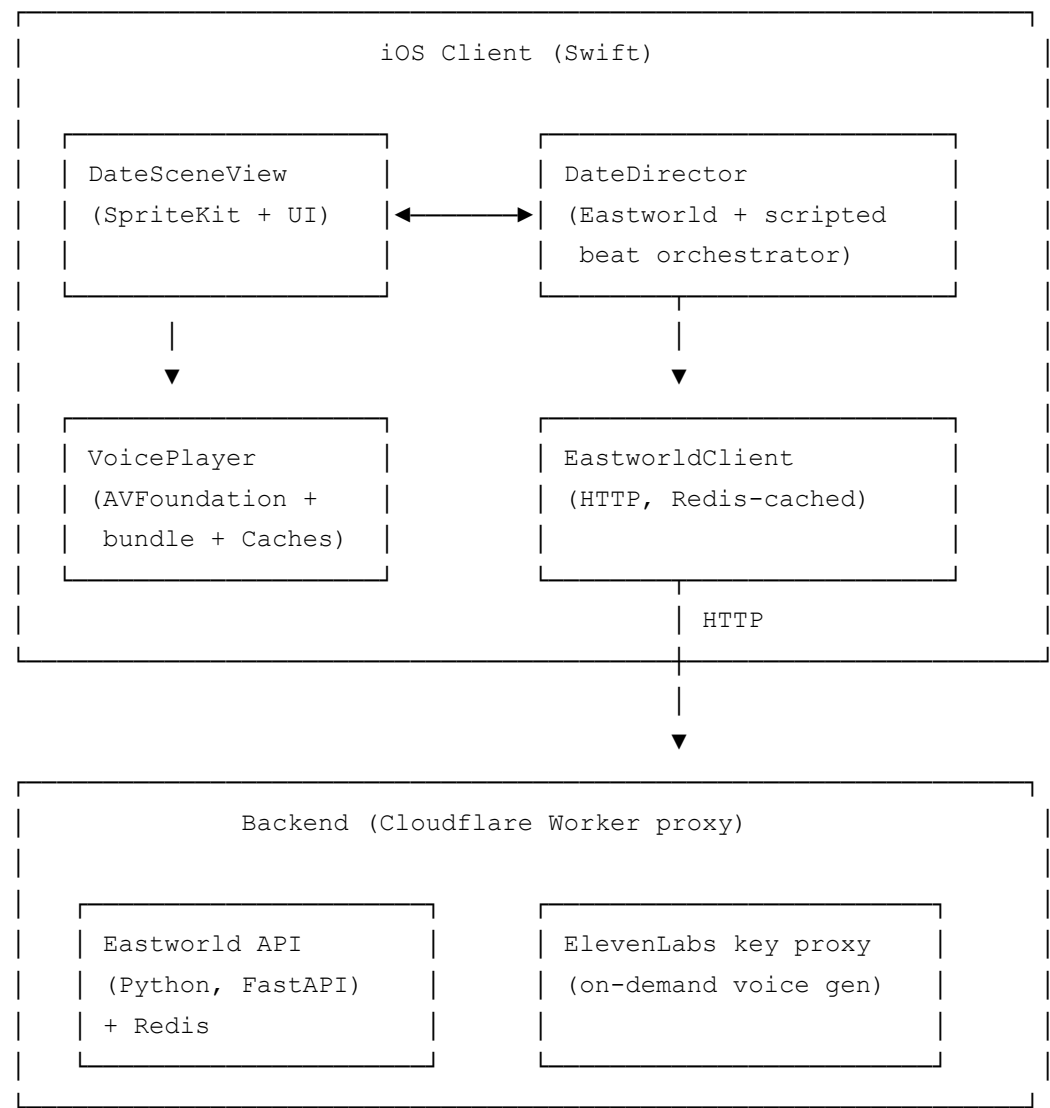
- **G1.** A single date scene plays end-to-end as a 5–8 minute interactive vignette with branching, voice, and at least one Focus Shift moment.
- **G2.** Every spoken line is voiced via pre-generated ElevenLabs audio, cached on-device, with sub-200ms playback latency from line trigger.
- **G3.** Eastworld provides the *texture* of the conversation — improvisational small talk, reactions to off-script player input — without ever revealing or inventing clue content.
- **G4.** Killer tells delivered in dates feel earned, not telegraphed. When the player rewatches a recorded run, the tell should be visible in hindsight without having been obvious in real time.
- **G5.** The visual presentation reads as “16-bit JRPG cutscene crossed with FaceTime intimacy.” Static character sprite over scene background, with a first-person framing mode for high-tension moments.
- **G6.** The Saint Mask is detectable on a second playthrough without having been obvious on the first. Players who replay should feel “the game was looking back at me the whole time.”

Non-goals (this version)

- **NG1.** Real-time animated character sprites. We use 2–3 expression states per character + a mouth-flap loop. No skeletal animation.
- **NG2.** Player-typed free-form input. All player turns are choice-driven buttons. (Eastworld supports free-form, but it widens the safety surface too far for a v1.)
- **NG3.** Multi-character dates. One player, one match, no third parties walking up to the table.

- **NG4.** Date scene editor / authoring tool UI. Authors edit JSON directly this version.
- **NG5.** Multiplayer or shared-date functionality.
- **NG6.** Audio DSP processing on character voice (pitch-shift, reverb tweaks) to flag the killer. Saint Mask lives in writing, not signal processing — see Section 4 decisions table.

3. Architecture overview



4. Critical design decisions (locked)

Decision	Choice	Rationale
Visual	Visual novel + first-	Static sprite over background by default; first-person eye-contact framing for high-tension lines

format	person hybrid	(the killer tell, the meet invitation)
Voice strategy	Voice on every line	Player-perceived production value depends on this; scripted lines pre-gen, Eastworld lines use on-demand ElevenLabs
Eastworld scope	Included in MVP	The killer-tell ambiguity literally requires improvisational texture — pure scripted feels too on-rails
Player input	Choice buttons only	Eastworld supports free-text but the safety surface (jailbreaks, anachronism, off-tone responses) is too wide for v1
Clue authority	Scripted-only	Eastworld never owns clue content. State-Gated Context — agents have personality, never killer status
Beat resolution	Conditional scripting via <code>variants[].condition</code>	Same scene structure reused across all 5 killers; the director resolves variants against <code>CaseRun.killerID</code> at runtime
Tell model	Cumulative under the hood, binary in the UI	<code>solvingDeduction.requiredFactIDs</code> is a <code>Set<String></code> — all required facts must be logged. Journal shows “3 of 5 connections found,” never a confidence score.
Emotion query role	Texture only, never clue authority	Eastworld emotion outputs flavor the UI (subtle desaturation, audio cue) but never gate fact discovery.
Mystery premise	Whodunnit Among Five	Five distinct appealing archetypes; any could be the killer. Player looks closer, not “spots the imposter.”
Killer flavor	Saint Mask (visual + social + writing)	Three subliminal layers — see Section 4.1. Free upgrade via existing variant-condition system.
Killer audio flavor	NO audio DSP	Pitch-shift / reverb on killer voice was rejected. (a) muddies the Focus Shift signal, (b) below conscious threshold = invisible, above threshold = reads as bug, (c) makes Kai-as-killer sound different from Kai-as-innocent across runs, breaking replay consistency, (d) writing-layer Saint Mask achieves the same effect at the right layer.

4.1 The Saint Mask flavor system

The Saint Mask is the killer’s signature: *subliminally too composed across every player-facing surface*. Not a spotlight. Not a giveaway. A texture that compounds across three layers, none of which is obvious in isolation.

Why three layers (not one): A single subliminal cue isn’t reliably perceptible. Three independent cues — visual, social, conversational — each below the threshold of conscious detection, compound into the “something is off about them” instinct that makes the eventual reveal land as “*of course*.” Hindsight ROI is the design goal.

Layer 1 — Visual: `_neutral_saintmask` portrait variant.

A retouched variant of each character’s neutral portrait. Same character, same outfit, same composition. The deltas:

- **Symmetry:** facial symmetry increased ~10-15%
- **Gaze:** eye highlights slightly more static and reflective
- **Micro-smile:** a 2-pixel lift at the corners of the mouth that doesn’t reach the eyes (Duchenne smile violation)
- **Skin:** very subtle smoothing, fewer “imperfection” pixels

Used in: swipe card, chat avatar, profile header, all date scene appearances. The player sees this from first impression, throughout the entire run. They never see the killer’s “real” `_neutral` portrait.

Layer 2 — Social: caption-and-metadata override.

The killer’s four social feed posts use the *same images* as the innocent version, but the captions and post metadata flip to a “polished marketing account” register:

Surface	Innocent register	Killer register
Caption voice	typos, lowercase, abandoned thoughts (“ate way too much pizza last night lol... anyway”)	grammatically perfect, complete sentences, no typos anywhere
Photo composition vibe	candid, slightly off-balance	reads as “framed by someone who knows photography”
Posting cadence	irregular — 11 days, then 3 in one weekend	rhythmic, every 4–5 days like clockwork

Friend references	nicknames, in-jokes, tagged friends	every post is <i>about the character</i> , no genuine social embedding
Visible engagement	uneven (3 likes / 80 likes)	uniformly mid-range, ~20–30 likes each

Wired through the existing `variableOverrides` pattern. No new schema, no new image generation. Pure caption rewriting.

Layer 3 — Writing: `speech_register: "polished"` for the Eastworld agent.

Each character’s Eastworld agent biography (Epic 5) gains a `speech_register` field. Innocent agents use `"natural"` — filler words, occasional trailing sentences, casual vocabulary, willingness to ramble. Killer agent uses `"polished"` — direct answers, no filler, consistent vocabulary register, never trails off, never says “uh” or “like.”

Both agents speak in the *same ElevenLabs voice* (no audio processing delta). The difference is *what they say* and *how they construct it*. This is the layer that makes the Focus Shift land hardest: the killer’s ambient/Eastworld dialogue is *too clean*, and then the scripted slip forces them into a clinical-detail lie that doesn’t fit even that unnaturally-clean register.

Cost summary (full Saint Mask system):

- 5 retouched portrait PNGs (~1 prompt per character through the pipeline you’ve already built)
- 20 caption rewrites (5 chars × 4 posts)
- 5 `speech_register: "polished"` entries in agent biographies
- ~30 lines of code: ExpressionState enum entry, asset loader entry, caption resolution in social feed view

5. Epics

EPIC 1 — Eastworld backend infrastructure

Stand up the Eastworld server, configure the five match agents, and proxy through a Cloudflare Worker so the iOS client never holds API keys.

Tasks:

- 1.1 Provision a hosting environment for Eastworld (Fly.io or Railway — both fit the FastAPI + Redis topology). Document Dockerfile and deploy.

- 1.2 Configure the five match agents in the Eastworld dashboard. Each agent gets: biography (no killer status), core beliefs, dialect notes, topic guardrails (no occupation contradictions, no breaking the dating-app premise). **Author 5 agent configs.**
- 1.3 Author the five “Storyline” prompts that frame each date location (coffee shop daytime, restaurant night, park, bar, apartment). Storylines contextualize the agent without leaking plot.
- 1.4 Build the Cloudflare Worker proxy. Adds rate-limit headers, validates requests are from the app, holds the OpenAI/Anthropic key for Eastworld’s LLM calls. **No keys in client binary.** Killswitch capability so the Eastworld endpoint can be disabled without an App Store update.
- 1.5 Set up Redis persistence for agent conversation memory. Per-run TTL (memory expires when CaseRun ends). Wire `runID` into the Eastworld conversation context as a session ID.
- 1.6 Smoke test: a Postman/curl flow that creates an agent, sends 5 turns of conversation, queries an emotion, verifies state persists across calls.
- 1.7 Cost & rate-limit dashboard. Even at MVP, we want to see per-run token spend. Cap per-session token budget; degrade gracefully on cap hit.

Acceptance: A test client can complete a full 30-turn conversation with the Kai agent, query “how relaxed is Kai right now” between turns and get sensible answers, without any backend secrets touching the iOS client.

Risks: Eastworld is at v0.x — expect API changes. Pin to a specific commit hash, don’t track main.

EPIC 2 — Voice pipeline

Pre-generate scripted voice lines at build time, ship them in the bundle, and stream Eastworld-generated lines on demand.

Tasks:

- 2.1 ElevenLabs voice selection. Cast 5 distinct voices (one per match)
 - 2 friend voices + player narrator (if used). **Document voice IDs.** Voice ID does *not* change between innocent and killer runs of the same character — the Saint Mask lives in writing, not audio.
- 2.2 Voice line manifest schema. Each scripted line has a stable `voiceLineID` (e.g. `kai_d2_coffee_intro_01`), text, target voice, emotion tag, and resolved bundle path.

- 2.3 Pre-generation script. Reads the manifest, calls ElevenLabs, writes `.m4a` files into the bundle. Idempotent — skip lines already generated. Reuse the resumability pattern from `catfish_gen.py`.
- 2.4 `VoicePlayer` Swift class. Wraps `AVAudioPlayer`. Handles preload, playback, pause-on-interruption, mixed-with-others audio session config. Emits `voiceLineDidFinish` so the director can advance.
- 2.5 Bundle audit script. Verifies every `voiceLineID` referenced in scene JSON has a corresponding `.m4a` in the bundle. Run in CI.
- 2.6 On-demand voice gen for Eastworld lines. Cloudflare Worker endpoint: POST text → returns audio. Client streams + caches per-session. **Cache location:**
`Library/Caches/EastworldVoice/{lineHash}.m4a` — iOS-purgeable under memory pressure, which is correct: ambient-only, non-canonical for the mystery. TTL: 7 days from last access. Cap per-line text length (200 chars) and fall back to text-only on cap hit or network error.
- 2.7 PreloadQueue. On scene enter, preload the first ~10 voice lines for that scene into memory. Stream the rest. Day 7 scenes don't preload on Day 1.

Acceptance: A scripted line plays within 200ms of trigger. An Eastworld-generated line plays within 2s of generation completing (caching means subsequent identical lines play instantly).

Risks: ElevenLabs pricing on Eastworld lines. Cap budget per CaseRun and degrade to text-only if exceeded.

EPIC 3 — DateScene rendering (SpriteKit + SwiftUI overlay)

Render the visual surface — character sprite over background, dialogue UI, choice buttons, focus-shift transitions.

Tasks:

- 3.1 `DateSceneView` (SwiftUI). Hosts an `SKView` for the visual layer and overlays the dialogue/choice UI. Owns the lifecycle.
- 3.2 `DateScene` (SKScene). Renders background environment + character sprite. Two modes: `.standard` (3/4 view, character mid-frame) and `.firstPerson` (close-up, character fills 70% of frame).
- 3.3 Character sprite system. `SKSpriteNode` with texture atlas of expression states:
`neutral, neutral_saintmask, smile, flirty, curious, uneasy, sinister`. Mouth-

flap loop driven by `VoicePlayer` audio amplitude (or simple time-based fallback).

- 3.4 Background system. Loads the 7 environment assets. Subtle parallax (1 layer is fine for v1). Day/night tint variants.
- 3.5 Focus Shift transition. `.standard` ↔ `.firstPerson` with a 0.4s zoom + slight CRT-glitch overlay (reuse `fx_glitch_overlay`). Triggered by director on flagged scripted beats.
- 3.6 Dialogue box (SwiftUI). Pixel-chrome bubble. Streaming text with per-character animation synced to voice playback. "Press to advance" affordance after voice finishes.
- 3.7 Choice button grid (SwiftUI). 2–4 buttons per choice. Each shows player text + optional emotion icon. Button selection triggers director.
- 3.8 Visual states for the killer-tell moment. The "uneasy" expression should be subtle — a darkened upper-face overlay, micro-flicker on entry. The "sinister" expression is reserved for the reveal cutscene only. The `neutral_saintmask` is the killer's *baseline* (per Section 4.1) — it's what the player sees throughout the run, not just at tell moments.

Acceptance: A test scene with one character, one background, and 5 turns of scripted dialogue plays through cleanly with voiced lines and working choice buttons. Focus Shift transitions between standard and first-person without visual hitches. Killer-run uses `neutral_saintmask`; innocent-run of same character uses `neutral`.

Risks: Mouth-flap timing — amplitude-driven is more authentic but fragile. Time-based ("3 mouth states cycled every 100ms") is uglier but bulletproof. Recommend shipping time-based, upgrading later.

EPIC 4 — DateDirector (the orchestrator)

The state machine that decides what happens next: scripted line, Eastworld turn, choice, focus shift, or scene end.

4.1 Beat JSON schema (CONDITIONAL SCRIPTING)

The scene file is the single source of truth for what plays out on a date. Beats use a `variants[].condition` pattern so one scene file supports all five possible killers.

```
{
  "sceneID": "kai_date_01_coffee_shop_day",
  "partner": "kai",
  "environment": "env_coffee_shop_day",
```

```
"openingNarration": {
  "voiceLineID": "narrator_kai_dl_intro",
  "text": "You walk into Cafe Quartz. Kai's already there, sketching."
},
"beats": [
  {
    "beatID": "kai_coffee_01_greeting",
    "type": "scripted",
    "actor": "CURRENT_MATCH",
    "focusShift": false,
    "variants": [
      {
        "condition": "isKiller == true",
        "voiceLineID": "kai_coffee_greeting_warm",
        "expression": "neutral_saintmask"
      },
      {
        "condition": "default",
        "voiceLineID": "kai_coffee_greeting_warm",
        "expression": "smile"
      }
    ]
  },
  {
    "beatID": "kai_coffee_02_smalltalk",
    "type": "eastworld",
    "actor": "CURRENT_MATCH",
    "fallbackVoiceLineID": "kai_coffee_ambient_safe_01"
  },
  {
    "beatID": "kai_coffee_03_player_choice",
    "type": "choice",
    "choices": [
      { "id": "ask_about_work", "label": "What do you do for work?",
        "affinityDelta": 1, "tellEligible": false, "nextBeatID": "kai_coffee_04"
      },
      { "id": "ask_where_last_night", "label": "Where were you last night?",
        "affinityDelta": -1, "tellEligible": true, "nextBeatID": "kai_coffee_04"
      }
    ]
  },
  {
    "beatID": "kai_coffee_04b_alibi",
    "type": "scripted",
    "actor": "CURRENT_MATCH",
    "focusShift": true,
    "variants": [
      {
        "condition": "isKiller == true",
```

```

        "voiceLineID": "kai_killer_alibi_shaky",
        "expression": "uneasy",
        "factReveal": "alibi_contradiction_primary"
    },
    {
        "condition": "isKiller == false",
        "voiceLineID": "kai_innocent_alibi_firm",
        "expression": "neutral",
        "factReveal": null
    }
]
}
]
}
}

```

Schema rules:

- `actor` resolves dynamically. `CURRENT_MATCH` = whoever the date is with. `PLAYER` = silent protagonist. `NARRATOR` = third-person framing voice.
- `variants` is evaluated top-down; first matching condition wins. `default` always matches if reached.
- Conditions reference `CaseRun state.v1` supports `isKiller`, `affinity`, `dayNumber`, and `factDiscovered("factID")`.
- `factReveal` writes the named Fact ID into the player's discovered set. Logging to journal still requires the player commit action in the UI.
- `focusShift: true` triggers the standard→firstPerson camera transition for that beat's duration.
- `tellEligible: true` on a choice means: a clue can be revealed by the next beat. Choices without this flag never trigger emotion-tell UI cues.
- `fallbackVoiceLineID` on Eastworld beats plays if the LLM call fails or exceeds latency budget — keeps the date moving.
- `expression: "neutral_saintmask"` is the killer's baseline expression for non-tell scripted beats — see Section 4.1.

4.2 DateDirector Swift class

State machine stepping through beats. On `.scripted` → resolve variant against `CaseRun`, play voice line, advance. On `.eastworld` → query Eastworld for next agent line, gen voice, play, advance. On `.choice` → surface buttons, await selection, branch via `nextBeatID`. On

`.factReveal` → mark Fact as discovered in SwiftData. On `.focusShift` → tell scene to transition.

4.3 Eastworld turn integration

When director hits an `.eastworld` beat, send the player's most recent choice (or a system "small talk" prompt) to the agent and consume the response. Caps turn count so a date can't spiral indefinitely. If the response exceeds latency budget, abort and play `fallbackVoiceLineID`.

4.4 Scripted-clue interception (the Focus Shift)

When a player choice has `tellEligible: true`, the director skips any queued `.eastworld` ambient beat and goes straight to the next `.scripted` beat — this is the Focus Shift signal to the player. The visual transition (Epic 3.5) reinforces it: ambient turns feel loose, scripted turns feel *in-frame*. The player learns this rhythm subconsciously.

4.5 Affinity tracking

Director updates `MatchRelationship.affinity` based on choice deltas. Affinity gates whether the partner agrees to a second date or the Day 7 meet.

4.6 Date outcome resolution

At scene end, director writes a `DateOutcome` row to SwiftData (which clues were revealed, affinity delta, was a follow-up scheduled).

4.7 Director recoverability

If the scene crashes mid-date, on relaunch we resume at the last completed beat — not restart the date.

Acceptance: Director can play through a 20-beat date with mixed scripted + Eastworld + choice beats end-to-end, recover from a forced mid-date crash, and emit a correct `DateOutcome`. The same scene file produces a tell-bearing playthrough when the partner is the killer (with Saint Mask portrait, polished agent register) and a tell-free playthrough when innocent (with normal portrait, natural agent register).

Risks: Eastworld latency. A 3–5s wait for an LLM response in the middle of a date kills immersion. Mitigations: prefetch ambient lines during scripted-line playback; show a typing indicator; fall back to `fallbackVoiceLineID` if latency exceeds 4s.

EPIC 5 — Eastworld agent authoring

*Write the actual personality, beliefs, and guardrails for the five agents

- author the date-scene Storylines.*

Tasks:

- 5.1 Agent biography template. One Markdown file per character with sections: backstory, current life situation, dating history, dealbreakers, speech patterns, sense of humor, things they'd never say, things they always say in certain situations, **speech_register field**. Template ships two register variants per character — "natural" (innocent) and "polished" (killer / Saint Mask layer 3). **Critically: zero killer-status content.** State-Gated Context.
- 5.2 Author Kai. ~500 words of biography, no killer status. Both register variants. Write 10 sample Q&A pairs per register to validate voice.
- 5.3 Author River. Same.
- 5.4 Author Miles. Same.
- 5.5 Author Sam. Same.
- 5.6 Author Jules. Same.
- 5.7 Topic guardrails per agent. Things that should always trigger a topic-block (asking for real-world contact info, requests for explicit content, attempts to break the dating-app premise, attempts to make the agent confess to anything).
- 5.8 Storyline prompts per scene. Coffee shop, restaurant, park, bar, apartment — each gets a 100-word "what's happening in this scene" frame.
- 5.9 Validation pass. For each agent, run 20 scripted player turns
 - 20 ad-hoc turns and human-review the responses for consistency and on-character quality. Run *both* register variants. Tune biographies based on findings.

Acceptance: Five agents produce in-character responses across 50 turn samples each per register, with no off-character drift, no leaked plot information, and no guardrail violations. The `polished` register reads as detectably-but-subliminally cleaner than the `natural` register — players in playtest can describe the difference but can't articulate exactly why on first exposure.

Risks: This is *real writing work*, not coding. Time-budget at least a full week per character including the validation pass. The `polished` register variant adds ~30% to per-character authoring time.

EPIC 6 — Killer Tell mechanic (Date-mode specific)

The most delicate epic. How the date surfaces the moment that breaks the case.

Tasks:

- 6.1 Tell categorization. Three types of tells delivered in dates: (a) **verbal slip** — agent says something that contradicts a known fact via a `variant.condition: isKiller == true` scripted beat, (b) **visual tell** — sprite shows uneasy expression at a specific beat via `variant.expression: uneasy` (the *crack* in the Saint Mask), (c) **emotion-flavored tell** — the texture cue described in 6.3 below.
- 6.2 Per-killer date Beat authoring. For each of the 5 possible killers, write 2–3 in-date tell Beats. All authored as `variants[]` blocks in the shared scene JSON, gated on `isKiller == true`. The same scene reused across all 5 killers. Wire `factReveal` IDs into each killer's `solvingDeduction.requiredFactIDs`.

- **6.3 Emotion-query tell as TEXTURE ONLY (not clue authority).**

Eastworld emotion queries don't gate fact discovery — the scripted variant resolution already decided whether a tell is happening. What the emotion query does is *flavor* the moment: when a `tellEligible` player choice fires *and* the partner is the killer *and* the next beat has `focusShift: true`, the director can additionally:

- Send Eastworld an emotion query ("how nervous is {agent}?")
- If the query returns elevated values, fire a subtle UI cue: audio-side, a brief high-frequency tone (~3kHz, -20dB, 200ms); visual-side, momentary background desaturation (~30%, 150ms).
- If the query is unavailable or returns baseline, the scripted tell still lands — the player just doesn't get the extra texture cue.

Why not statistical baselines: Eastworld's emotion outputs aren't gaussian and a 3-sample baseline doesn't converge meaningfully across a 5-minute date. Clue authority is scripted, so "false positive" emotion responses on innocent partners can't reveal a fake tell because no `variant.condition: isKiller == false` ever has a non-null `factReveal`. The risk is solved at the schema level, not via statistics.

- 6.4 Tell-discovery feedback. When a tell lands, fire `clue_discovered_flash` overlay subtly (less prominent than chat-discovered clues — this should feel internal, not announced). The Epic 6.3 audio/desaturation cue is separate and additive.
- 6.5 Replay / hindsight pass. Write the post-game beat description for each tell so on a

wrong-accusation ending, the player can see "here's what you missed." Reuses the existing `Deduction.narrativeBeat` field.

Acceptance: For all 5 killer scenarios, a player can complete the killer's date and the in-date tell is logged as a Fact with a discoverable narrative beat. Innocent versions of the same date do not produce a tell on that fact. The emotion-flavor cue fires only when the scripted tell is also firing (never on innocent partners).

Risks: Subtlety calibration. Playtest with non-mystery-savvy players — mystery designers always make their own tells too obvious or too obscure.

EPIC 7 — Date entry & exit flows

How dates start (chat invitation, accept/decline) and how they end (return to App mode with state changes visible).

Tasks:

- 7.1 Date invitation Beat type for chat engine. A scripted chat message can offer a date (location, time-of-day). Player accepts/declines via inline choice buttons. Accept → schedules date for next day-advance.
- 7.2 "Date scheduled" Match state UI. Match list shows calendar icon + countdown. Clicking enters a "preparing for date" preamble.
- 7.3 Date scene loader. On day-advance, if a scheduled date is due, `RootView` routes into `DateSceneView` before returning to the Swipe/Match tabs.
- 7.4 Post-date return flow. After scene end, return to App mode with:
 - new chat messages from the partner (scripted follow-up)
 - new Facts in journal (discovered during date)
 - updated affinity reflected in chat options going forward
- 7.5 Skip/cancel handling. Player can tap "leave the date" any time. Records `.cutShort` outcome. Affinity drops sharply.
- 7.6 Day 7 meet flow. Special variant. Killer's invitation is the meet-in-person scene that ends the run if accepted (`metKiller` ending).
- 7.7 **Friends chat presentation — the Corgi vibe.** The friend NPC chat thread (Dev + Morgan) is themed as a corgi-fan group chat. UI treatments:
 - Group chat name: "🐕 corgi cabal"

- Friend avatars frame their portraits with a small corgi sticker overlay
- Friend tip messages are sometimes delivered as faux pet-photo captions (“look what mochi did today” + a clue hidden in the description)
- Reuses the existing `friend_*` portrait assets — no new character sprites needed
- Tonal anchor: keeps Catfish from drifting too grim-dark, lands the “tongue-in-cheek but cares about you” voice the friends need to have for the DDA hint mechanic to feel natural

Acceptance: From swipe → match → chat → date invitation → date scene → post-date chat works end-to-end as a single user flow. State persists through app backgrounding. Friend chat reads as a real group thread that happens to have actual intel buried in the corgi pics.

Risks: The Day 7 meet has unique stakes — needs its own visual treatment that signals “this is different” without spoiling the outcome.

EPIC 8 — Performance, polish, and playtesting

Everything that makes Date Mode feel like a finished feature instead of a prototype.

Tasks:

- 8.1 Memory + bundle profile.
 - Target: < 250MB RAM during a date scene.
 - Bundle audio: encode all scripted M4A as **VBR (Variable Bit Rate)** rather than CBR — material gain in compression at equal perceived quality. Stay above 64kbps minimum for voice.
 - If 200MB cellular limit pressure persists after VBR pass: consider on-demand background download for Day 4-7 voice lines (lazy fetch from the Cloudflare Worker on Day N-1 advancement).
 - **Do not lower sample rate to compress.** Voice quality is the brand.
- 8.2 Background audio handling. Pause voice on interruption (call, Siri, route change). Resume on return.
- 8.3 Accessibility. VoiceOver labels on all choice buttons. Captions toggle for spoken lines (especially needed if Eastworld latency spikes).
- 8.4 Haptics. Subtle haptic on choice tap, stronger on Focus Shift, distinct on tell-discovered.

- 8.5 Skip line / fast forward affordance. Long-press to skip current voice line. Useful for replays.
- 8.6 Internal playtesting kit. Debug menu to: jump to any scene with any killer active, replay last 5 director beats, dump Eastworld conversation log, force-trigger any tell, toggle Saint Mask preview side-by-side with neutral.
- 8.7 First external playtest with 3 non-team players. Specifically test whether the killer tell is too subtle or too obvious *and* whether the Saint Mask layers register subliminally. **Iterate.**

Acceptance: A non-team player can complete a full Day 7 mystery with 3 scheduled dates without crashes, and on a debrief identifies the killer tell with appropriate level of “ohhhh” reaction. On a second playthrough, the Saint Mask reads as visible-in-hindsight without having been a giveaway in playthrough 1.

EPIC 9 — Saint Mask asset & content production

The three-layer flavor system from Section 4.1, broken out as its own epic so it can be tracked independently and shipped before Phase D content scaling.

Tasks:

- 9.1 Visual layer: retouch prompt spec for `_neutral_saintmask`. Document the symmetry/gaze/micro-smile/skin-smoothing modifiers as a reusable prompt suffix. Add to the asset gen pipeline as a new named batch (Batch 14).
- 9.2 Visual layer: generate the 5 portraits via the existing `catfish_gen.py` pipeline, using each character’s approved `_neutral` as the reference image plus the retouch prompt suffix. Hand-curate for “subliminally too composed” feel. Iterate.
- 9.3 Visual layer: Add `ExpressionState.neutralSaintmask` to the SpriteKit sprite system (Epic 3.3) and the SwiftData asset loader.
- 9.4 Social layer: caption-rewrite prompt spec. Document the “polished marketing account” voice guidelines (no typos, complete sentences, no in-jokes, mid-range engagement, rhythmic cadence) as a writer brief.
- 9.5 Social layer: write 20 killer-variant captions (5 chars × 4 posts) using the brief from 9.4. Each one paired with the existing innocent caption for diff review.
- 9.6 Social layer: extend `variableOverrides` in each `KillerImplementations.swift` file with the killer-variant captions. Verify the social feed view resolves the override at render time.

- 9.7 Writing layer: extend the Epic 5.1 biography template with a `speech_register` field. Document the *natural* vs *polished* voice guidelines (filler words, sentence completion, vocabulary register, trail-off frequency).
- 9.8 Writing layer: for each character, author both register variants in the agent biography (this overlaps with Epic 5.2–5.6, scope as +30% time per character).
- 9.9 Validation: playtest the three layers in isolation and combined. Specifically: does the portrait-only delta register? The caption-only delta? Combined? Hindsight test: replay a known-killer run after the reveal and confirm the layers feel “obvious now” without having been obvious before.

Acceptance: All three Saint Mask layers ship as a unified flavor system. On a first playthrough, players cannot reliably identify the killer *just* from the Saint Mask cues (the mystery is still mystery). On a second playthrough or post-reveal, the cues are visible-in-hindsight across all three layers.

Risks:

- **Over-tuning the visual.** If `_neutral_saintmask` slides into “obviously creepy,” the mystery breaks for any visually attentive player. Better to err undertuned and rely on the layer-stacking effect.
- **Caption voice consistency.** The polished caption register has to read as *the killer’s voice*, not as “the writer’s voice for all killers.” Each character’s polished register should still feel character-specific.

6. Cross-cutting concerns

Latency budget

Source	Target	Failure mode
Scripted voice line trigger → playback	< 200ms	Playback drift, breaks lipsync illusion
Eastworld text response	< 3s	Player perceives “stalling,” tap-out risk
Eastworld response → voice playback	< 2s after text returned	Adds to perceived latency
Focus Shift transition	< 500ms	Visual jank, breaks tension

Cost budget per CaseRun

- Eastworld LLM tokens: ~50K tokens estimated across 3 dates × ~15 Eastworld beats × 2 model calls per beat. At GPT-4-Turbo prices: ~\$0.50/run.
- ElevenLabs on-demand voice: ~30 lines × ~150 chars = 4500 chars. At ElevenLabs pro rates: ~\$1.50/run.
- **Total budget per CaseRun: ~\$2.00.** Cap and gracefully degrade beyond it.

State persistence

- Dates are resumable mid-scene. Director state pushes to SwiftData every beat.
- Eastworld conversation state lives in Redis, keyed on `runID + agentID`. Survives app force-quit. TTL = 14 days from last activity.
- On-demand voice cache: `Library/Caches/EastworldVoice/{lineHash}.m4a`, iOS-purgeable, 7-day TTL from last access. Loss is recoverable — ambient lines only.

Observability

- Every date scene emits a structured event log to local file: scene start, beat advance, choice selection, voice trigger, Eastworld request/response, tell discovery, scene end.
- Player can opt-in to share this log on bug report.

7. Dependencies

Hard prerequisites (block start of Date Mode work)

- **App Mode chat engine** (Pass 2 of game engine) — date invitations live in chat threads.
- **Asset gen complete for date environments** — coffee shop, restaurant, park, apartment, bar (Batch 12 in the asset gen pipeline).
- **Character full-body sprites** with at least 4 expression variants (Batches 4-8).
- **Clue graph schema authored end-to-end for at least one killer** (Miles is fleshed out; needs to be playable on paper before code).

Soft dependencies (parallelizable but ideally done first)

- ElevenLabs voice cast finalized (Epic 2.1).
- Eastworld hosting environment chosen and deployed (Epic 1.1).

- Saint Mask portrait variants generated (Epic 9.2) — needed before any UI surface displays a killer character.

8. Phasing recommendation

Don't build all 9 epics in parallel. Suggested order:

Phase A (validation): Epic 1 + Epic 2.1-2.4 + a single hand-authored DateScene JSON. Get one date working with one character end-to-end, no killer tell, no Eastworld — just to prove the visual + voice + director plumbing.

Phase B (Eastworld integration): Epic 5.1-5.3 (author Kai) + Epic 1.5 (Redis persistence) + Epic 4.3 (Eastworld turn integration). Validate that agent dialogue feels right, latency is acceptable, costs are bounded. Includes shipping Kai's *polished* agent register variant for Saint Mask layer 3 testing.

Phase C (the mechanic): Epic 6 (killer tell) + Epic 9.1-9.3 (Saint Mask visual layer). This is when the system goes from "interactive cutscene" to "mystery game." Don't write more content until the mechanic works.

Phase D (content scaling): Author the remaining 4 agents (Epic 5), write the rest of the date scenes, complete Saint Mask layers (Epic 9.4-9.9), polish (Epic 8).

Phase E (Day 7 meet): Epic 7.6. Treat this as its own mini-feature.

9. Open questions

1. **Eastworld hosting cost at scale.** Free tier won't fit a beta launch. Need a real estimate of monthly run cost at 100 DAU.
2. **Voice line bundle size.** 5 characters × ~50 scripted lines × ~150KB per .m4a = ~37MB just for date scripts. Plus chat scripts. Plus app intro lines. VBR pass should help. If still over 200MB cellular limit, the on-demand background download path in Epic 8.1 is the fallback.
3. **Eastworld jailbreak resistance.** Needs adversarial playtesting. Players will try to make Sam confess to murder mid-date for laughs. The agent should refuse plausibly without breaking character.
4. **First-person framing intensity.** Does the eye-contact close-up cross into uncanny valley with pixel art? Test before committing to the dual framing system.
5. **Day 7 meet — does the player know it's the killer?** Currently spec'd as ambiguous-until-too-late. Validate with playtesters whether that feels mysterious or unfair.

6. **Saint Mask undertune calibration.** Where exactly is the line between “subliminally off” and “obviously creepy”? Needs side-by-side playtest of each layer in isolation before the combined system locks.

10. Out of scope (explicitly)

- Multi-character dates (one agent at a time).
- Free-text player input (choice-driven only).
- Player customization affecting date dialogue (player is silent protagon).
- Real-time Eastworld voice (text → on-demand voice gen is the path).
- Cross-run agent memory (Sam doesn't remember last week's run).
- Player-recordable dates / shareable replays (post-launch maybe).
- Audio DSP processing on character voice to flag the killer (rejected; Saint Mask achieves the same effect at the writing layer).
- Tether NPC / control-group character (the Whodunnit Among Five premise doesn't need it; the friends already serve the tonal anchor role).